

MTDSE_CLUS_Batch10_SEM 2_1-2023_EC3R_DSECLZG529_Data Management for Machine Learning_Online

SANTHOSH A | 14 Apr 2024



Overall Status: Completed

Detailed Status: Time Over

Test Finish Time: April 14, 2024 11:30:01 AM IST

SA

SANTHOSH A

2022DC04445@WILP.BITS-PILANI.AC.IN

Campus ID: 2022DC04445

Test-Taker ID: - 131148327

Profile Picture Snapshot



Identity Card Snapshot



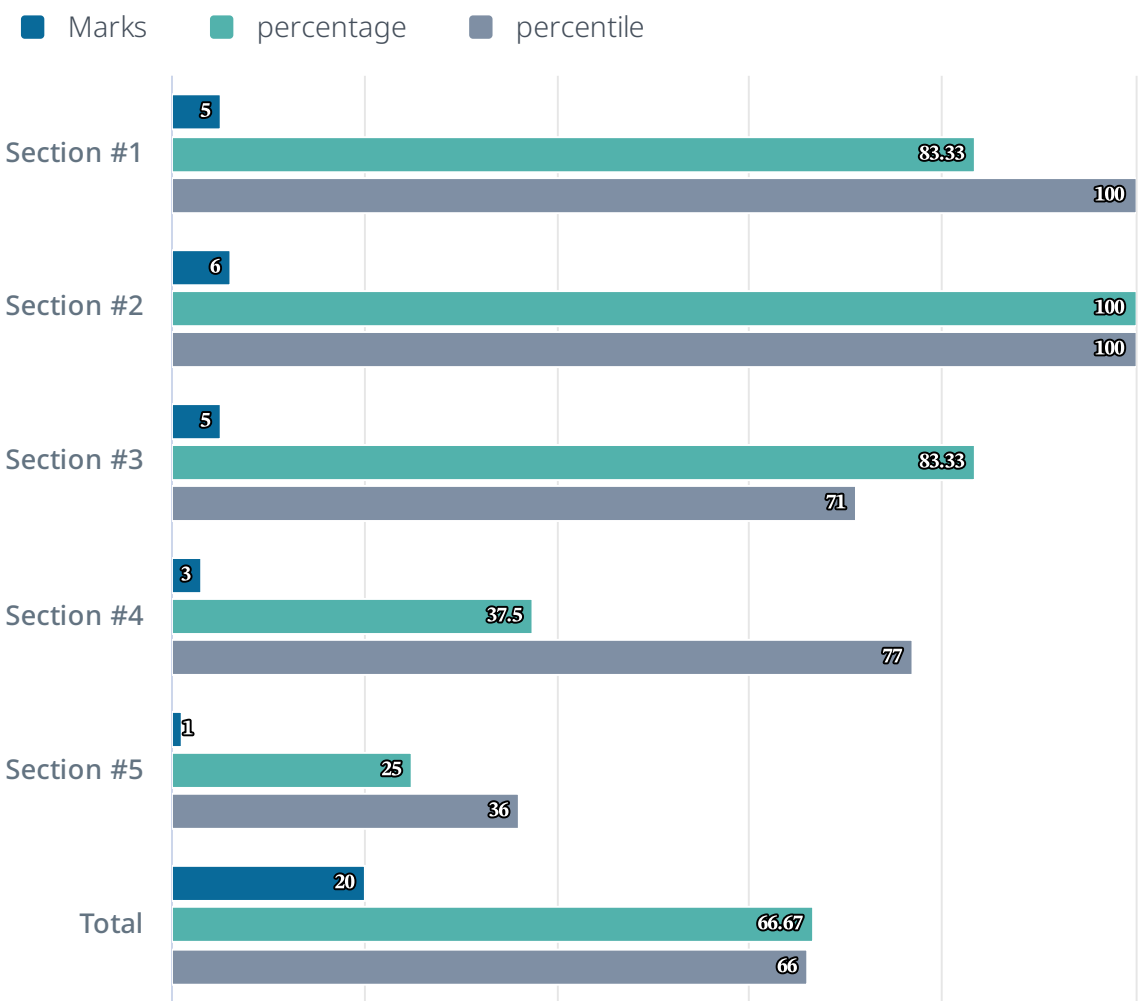
Overall Summary

20 Marks Scored
out of 30

66.67 % 66.31 percentile
out of 282 Test
Takers

2_h 29_m 53_s Time taken
of 2hr 30mins

Marks Scored



Attempt Summary

Distribution of questions attempted in a total of 5 question(s).

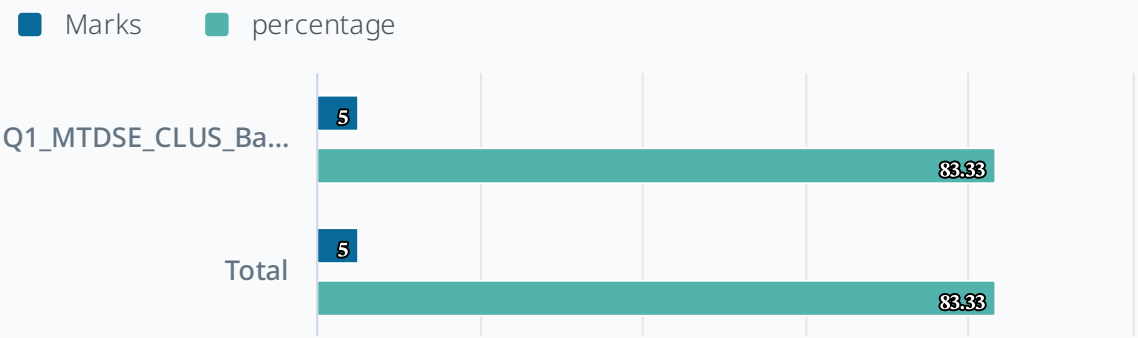


Correct	1 Ques	6/6 Marks
Incorrect	0 Ques	0/0 Marks
Partially Correct	4 Ques	14/24 Marks
Not Attempted	0 Ques	0/0 Marks

Section-Wise Details

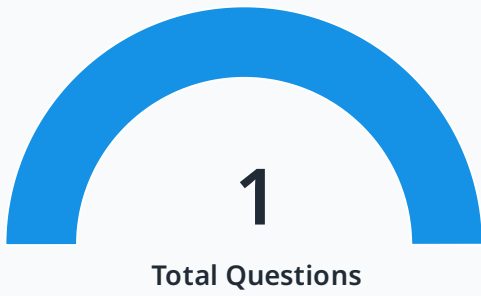
▼ Section 1	question(s)	Time taken	Marks Scored
Section #1	1 Q.	28m 6s (Untimed)	5 / 6

Marks Scored



Attempt Summary

Distribution of questions attempted in a total of 1 question(s).



This shows the correctness of questions attempted by the test taker

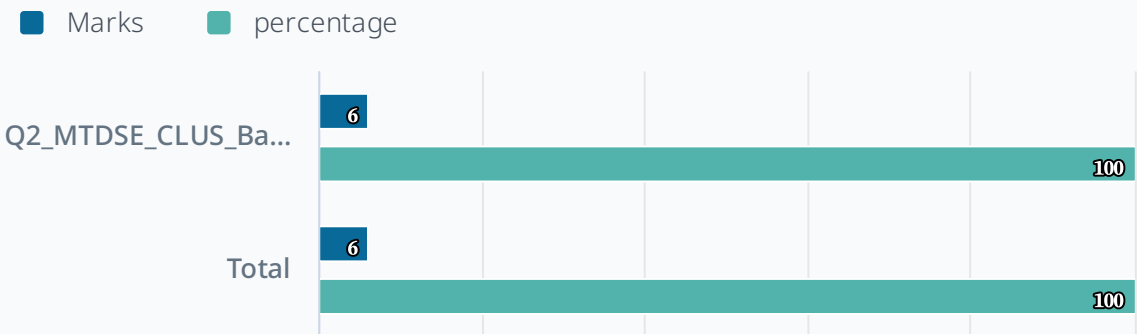
■ Partially Correct	1 Ques	5/6 Marks
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Section-Wise Details

Section 2
Section #2

question(s)	Time taken	Marks Scored
1 Q.	34m 34s (Untimed)	6 / 6

Marks Scored



Attempt Summary

Distribution of questions attempted in a total of 1 question(s).



Total Questions

This shows the correctness of questions attempted by the test taker

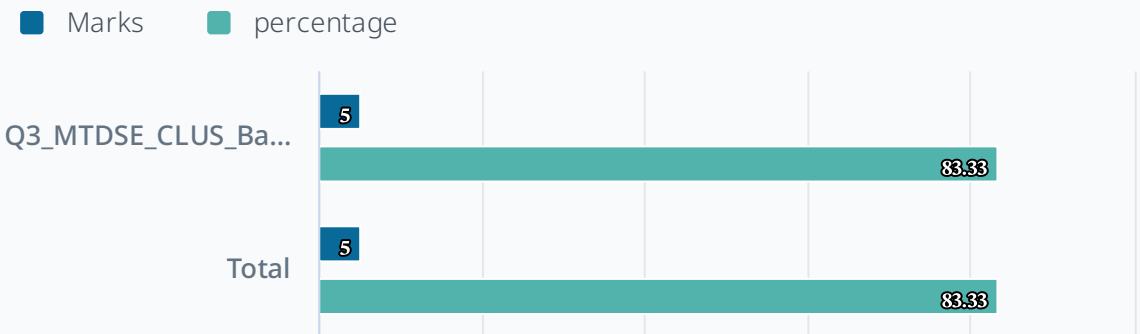
■ Correct	1 Ques	6/6 Marks
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Section-Wise Details

Section 3
Section #3

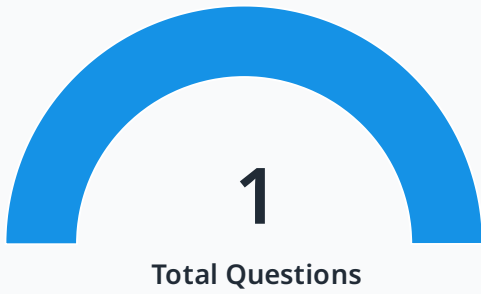
question(s)	Time taken	Marks Scored
1 Q.	33m 2s (Untimed)	5 / 6

Marks Scored



Attempt Summary

Distribution of questions attempted in a total of 1 question(s).



This shows the correctness of questions attempted by the test taker

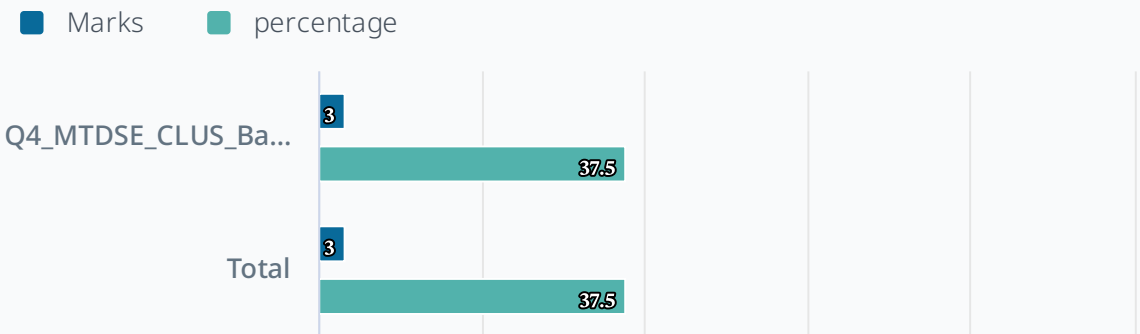
■ Partially Correct	1 Ques	5/6 Marks
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Section-Wise Details

Section 4
Section #4

question(s)	Time taken	Marks Scored
1 Q.	33m 46s (Untimed)	3 / 8

Marks Scored



Attempt Summary

Distribution of questions attempted in a total of 1 question(s).



Total Questions

This shows the correctness of questions attempted by the test taker

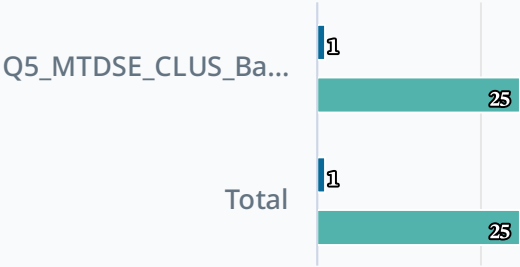
■ Partially Correct	1 Ques	3/8 Marks
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Section-Wise Details

Section 5	question(s)	Time taken	Marks Scored
Section #5	1 Q.	20m 23s (Untimed)	1 / 4

Marks Scored

Marks percentage



Attempt Summary

Distribution of questions attempted in a total of 1 question(s).



Total Questions

This shows the correctness of questions attempted by the test taker

Partially Correct 1 Ques 1/4 Marks

Q.
1

▼ Question 1

⌚ Time taken: 28m 6s

Marks Scored: 5/6

Question

Please answer this question only by typing your answers in the typing space available for this question

You are part of a data science team working for a large retail corporation that is expanding its operations to include online sales. Your team is responsible for developing machine learning models to optimize inventory management and predict customer demand for various products. One crucial aspect of your work is managing and versioning the data used to train and test these models.

Imagine you have been tasked with designing a data versioning system for the retail corporation's machine learning projects. How would you approach this task? Outline the key components and considerations you would consider when implementing the data versioning system. [6 Marks]

Response

DVC system

A good data version control tool will allow to have a unified datasets with strong repositories to conduct any of the ML experiments

DVC tools are very critical ,if we care about tracebility, reproducability and ML model lineage

To choose appropriate DVC tool points to be consider are

- 1. Support for data modalarity
- 2.Ease of use
- 3. Diff and compare capability
- 4.how well it work with the stack
- 5. Is team ready for onboarding

The best DVC tool avaialble in markets need to be explored for the current needs. The currently best tools in the market are

- 1. Neptune
- 2.Pachyderm
- 3. DVC
- 4. Glt Large file storage
- 5. LakeFS
- 6. Delta Lake

Things to be considered while setting up DVC tools

- 1. How easy to setup
- 2. Availability of the team to setup
- 3.What tool stack currently using today

4. Data modality.

Key components for DVC Systems :

1. Data Storage : Use any centralized cloud storage soulike like Google cloud storage, Amazon S3 or any other tool which allows scalability and accessibility

2. Version control system (VCS)

Implementation o VCS like git to track any changes made to data files

Integrate the VCS with the data storage solution for seamless version management. Tools like DVC can facilitates this migration

3. Metadata management :

Store the detailed metadata for each data version including

- Version number and date creation
- Change logs/description
- Data pipeline information
- model performance metrics with each of the data version

Utilize a dedicated metada to store RDBMS for structed data and apache atlas for semi-structure or unstructed data

4. Data lineage tracking :

- Implement tools to tract the origin and transformation steps applied to each data version which allows tracebility and impact of data changes to model
- tools like apache airflow will enable to visualzie data pipelines and their dependencies

5. Access Control :

Define any needed users roles and permission for accessing and modifying the data versions which ensures data integrity and prevents unauthorized access

Words : 325

EVALUATOR'S NAME	MARKS ⬆	EVALUATION COMMENTS	LAST MODIFIED ON ⬆
Lokesh S	5.0	Designing a Data Versioning System for Retail Corporation's Machine Learning Projects: ***** ***** ***** Key Components: [2 marks] • Version Control System: Implement a version control system like Git to manage data versions efficiently. Each dataset should be treated as a repository, allowing for tracking changes, branching, and merging. • Metadata Management: Develop a metadata repository to store information about each dataset version, including attributes like creation date, modifications, contributors, and associated machine learning models. ***** ***** ***** Key Considerations: [4 marks] • Data Versioning Policy: Establish clear guidelines for when and how to create new versions of datasets. This policy should outline criteria for version increments, such as significant schema changes, data updates, or improvements in data quality. • Data Schema Evolution: Implement strategies to handle schema changes across different dataset versions. Consider using techniques like schema evolution tools or schema-on-read approaches to accommodate evolving data structures without disrupting existing machine learning pipelines. • Data Drift Detection: Incorporate mechanisms to monitor data drift over time. Utilize statistical methods or machine learning algorithms to detect changes in data	05:53 PM Apr 24, 2024

		distributions, ensuring that models remain accurate and up-to-date. • Reproducibility Measures: Implement procedures to ensure reproducibility of experiments across different data versions. This involves documenting all pre-processing steps, feature engineering techniques, and model configurations used for each experiment.	
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Q.

1

▼ Question 1

🕒 Time taken: 34m 34s

Marks Scored: 6/6

Question

Please answer this question only by typing your answers in the typing space available for this question

You are working as a data engineer for a healthcare technology startup that specializes in developing machine learning models to assist medical professionals in diagnosing diseases from medical imaging data. Your team is responsible for designing and implementing data pipelines to pre-process, label, augment, and validate the imaging datasets used to train and evaluate these models. Imagine you have been tasked with designing a comprehensive data engineering and ML engineering pipeline for the healthcare startup's medical imaging project. [1 + 1 + 1 + 1 + 1 + 1 = 6 Marks]

- a) How would you approach the data labelling process for medical imaging datasets, considering the critical importance of accurate annotations for training machine learning models?
- b) What techniques or tools would you use to efficiently label large volumes of medical images while ensuring high-quality annotations?
- c) Explain the significance of data augmentation in improving the generalization and robustness of machine learning models trained on medical imaging data.
- d) Outline common data augmentation techniques suitable for medical imaging datasets, considering factors such as image modality (e.g., X-ray, MRI, CT scan) and anatomical variability.
- e) Describe the importance of data validation in ensuring the quality and reliability of training and evaluation datasets for machine learning models in healthcare applications.
- f) Outline the process of data validation, including techniques for detecting and handling data anomalies, outliers, and inconsistencies in medical imaging datasets.

Response

- a) crowd-sourcing platoforms : Identify the platforms where qualified midical professional for labelling task. Platforms such as Hivemind etc
- In-house labelling team: Hire a medical professional in-house for labelling the data
- Hybrid method : Combining both crowd-sourcing and in-house team, which could be a cost-effective.
- b) Tools for labelling large volumes of medical images
- Collabrative labelling tools
- Double annotation
- Active learning
- automated-pre annotation tools
- c) Data augmentation artifiically expands the traning dataset by generating the variations of the existing images which helps
- Improvised model genaration : as wide range of data variations are there which results is best-fit and do better performance
- Increase model robustness : Model will best handle any noise
- d) Data augumentation:

For image-level techniques (applicable for any industry)

- Random flipping either horizontal or vertical flipping
- Random rotation
- Random crop or zoom operations
- Adding a noise manually
- 'variations in image quality like constrast, brightness etc

Modality-specific technique

- Simulate with different x-ray beam machnies and sensitivity
- MRI : provide different slice samples from different sources and different MRI machines
- CT Scan : Introduce any intentional imperfection data

e) Data validation is a key factor for model performance which ensures the quality and reliability of the training and evaluation of the datasets

1. Improve model performance
2. Reduce any biases or errors
3. Increase transparency

f) Data validation process involves the following major activities

- Data quality checks : Identify any missing values , inconsistency data, data-format validation and any inconsistency between lables and images
- Outlier detection: Using any of the statistical methods like IQR or any other method need to identify any of the outliers
- Label verification : Ensure the labelled images are correctly labelled and is consistent with the all the annotators
- Visual inspection : Identify any artifcats, noise or unexpected anatomical variations in the labelled images
- Exper review: Review any anomalies with the experts in the medical fields

Words : 298

Max marks: 6			
EVALUATOR'S NAME	MARKS ⬆	EVALUATION COMMENTS	LAST MODIFIED ON ⬆
Apoorva MS	6.0	.	11:55 PM Apr 17, 2024

Q. 1

▼ Question 1

🕒 Time taken: 33m 2s

Marks Scored: 5/6

Question

Please answer this question only by typing your answers in the typing space available for this question

Consider a scenario where a data analytics platform is being developed for a car rental company. The platform aims to analyze rental transactions and customer interactions to optimize operations and improve customer satisfaction. The data warehouse consists of three dimensions: time, vehicle, and customer, and the two measures are count (number of rentals) and revenue (total revenue generated from rentals). [1 + 2 + 2 + 1 = 6 Marks]

- a) Enumerate two classes of schemas popularly used for modeling data warehouses in the context of the car rental company. [1 Marks]
- b) Draw a schema diagram for the above data warehouse using one of the schema classes listed in (a). [2 Marks]
- c) Starting with the base cuboid [day, vehicle, customer], what specific OLAP operations should be performed to analyze the average rental duration for each vehicle type? [2 Marks]
- d) To obtain the same analysis, write an SQL query assuming the data are stored in a relational database with the schema rental_transactions (rental_date, vehicle_type, customer_id, rental_duration, revenue). [1 Marks]

Response

- a) **Star Schema** : A fact table in the middle connected to a set of dimensions table
- Snowflake Schema** : A refinement to star schema where some dimensional hierarchery is normalized in to set of smalle dimensional table forming a shape similar to snow flake
- Fact constellation:** Multiple fact table shares the dimension table, views as collection of stars, called galaxy-schema, or fact constellation
- b) *Unable to show linkage between dimensional tables in this windows.*
- But a fact table in this diagram will be connected to 3 dimensional tables i.e., time_table, Vehicle_table, Customer table.
- Eech key entry (dimentsion entry) in the fact_table need to connect to appropriate table

fact table

time_key
vehicle_key
Customer_key
Number of rentals
Revenue

time_table (dimensional table)

time_key

checkin time & date
delivery time & date

vehicle_table (dimensional_table)

vehicle_key
Owner details
Insurance details
ODO meter reading
Previous rental details

Customer_table(dimensional_table)

customer_key
contact#
Address
Previous rental history
Driving license details

c) OLAP operations for average rental duration

Drill down : Starting with the base cuboid [day, vehicle, customer], we candrill down on the vehicle dimension to analyze average rental duration for each vehicle type. This require aggregation data by vehicle type while keeping day and customer details intact

Roll up : Roll-up to the day dimension to analye average rental for any given day for vehicle type in a given period . This requires aggregation data by monthly , yearly while maintaiing vehicle type info

d) SQL query

```
SELECT vt.vechicle_type, AVG(rental_duration) AS average_duration
FROM rental_transactions rt
INNER JOIN vehicle_v ON rt.vehicle_id = v.id
INNER JOIN vehicle_type vt ON v.vehicle_type_id = vt.id
GROUP BY vt.vehicle_type
```

For best view please see test taker response in web view. [Shared Report Link](#)

Words : 258

Max marks: 6			
EVALUATOR'S NAME	MARKS ↕	EVALUATION COMMENTS	LAST MODIFIED ON ↕
Santosh Singh	5.0	Incorrect query.	12:58 AM Apr 29, 2024

Question

Please answer this question only by typing your answers in the typing space available for this question

You have been assigned the responsibility of designing a robust data architecture for a rapidly growing social media platform. This platform witnesses a significant influx of user-generated content, interactions, and engagement on a daily basis. To ensure seamless operations and efficient data management, your data architecture must cater to real-time analytics, reporting, and scalable data storage solutions. Consider the following critical metrics for the social media platform:

- 50,000,000 user-generated posts per day
- 20,000,000 likes, comments, and shares per day
- 10,000,000 new user registrations per day

Your task is to design a comprehensive data architecture to handle and process this substantial volume of data efficiently. Your architecture should encompass the following components and provide an estimate of the required storage capacity:

- Data Ingestion Layer
- Data Storage Layer
- Data Processing Layer
- Data Analytics and Reporting Layer

- a) In your response, outline the key features and technologies you would incorporate into each layer of the data architecture to ensure optimal performance, scalability, and reliability. **[4 Marks]**
- b) Additionally, provide an estimate of the required storage capacity based on the given metrics, considering factors such as retention period, and redundancy requirements. **[4 Marks]**

Response

a)

Data ingestion Layer :

Features :

- Real-time streaming for capturing user-generated posts, like comments and new registrations
- High throughoupt to handle millions of events/sec
- Scalable and reliable data collections across different sources
- Fault tolerance with date replication to ensure data availability

Technologies : Kafka, Amazon kinesis

Data storage layer:

Features :

- Scalability & distributed databases for structured data
- Object storage for unstructured data such as multimedia content
- enable data partitioning and fast access and retrivel when needed
- Data replication
-

- **Technologies** : Cassandra, Google Bigtable, Amazon S3, Google cloud storage for unstructured data

Data Processing layer

Features :

- Distributed data processing to handle large size data and its transformation and aggregations
- To support batch, mini-batch processing in real-time
- recovery mechanism and fault tolerance capabilities for data consistency

Technologies : Spark, Flink, Google dataflow

Data Analytics and Reporting layer :

Features :

- Dynamic and realtime dashboards for monitoring the user engagement and activity tracking
- Advanced analytics capability for generating insights and trends
- Integration with data storage layer for accessing and visualizing the data
- Security and access controls for data privacy

Technologies : Tableau, Power BI

b) Assume

each post size = 1 KB

Avg size of comment, like, share = 0.5 KB

New user registration size = 0.1 KB

Calculations : Data size

POSTs /day = $5 \times 10^7 \times 1 \text{ KB} = 5 \times 10^7 \text{ KB}$

Interactions (comments, like, share) / day = $2 \times 10^7 \times 0.5 = 1 \times 10^7 \text{ KB}$

New User registrations / day = $1 \times 10^7 \times 0.1 \text{ KB} = 10^6 \text{ KB}$

Total data volume / day = (posts + interactions + new user registration) = $6.1 \times 10^7 \text{ KB / day}$

Retention period : assuming a retention of 5 years

Data storage requirement will be $6.1 \times 10^7 \times 5 \times 365 = 11.13 \times 10^{10} \text{ KB}$

Redundancy requirements : For data redundancy of 3 fold

data storage required will be $3 \times 11.13 \times 10^{10} \text{ KB}$

1GB = 2^{30} Bytes

Data storage requirements = 103679 GB (For 5 years)

With redundancy requirements = $3 \times 103679 \text{ GB} = 311038 \text{ GB}$ (For 5 years with 3 fold requirements)

Words : 303

Max marks: 8			
EVALUATOR'S NAME	MARKS ⬆	EVALUATION COMMENTS	LAST MODIFIED ON ⬆
Kamlendra	3.0	key features and technologies incomplete for each layer of the data architecture; Estiamtes are not correct for all metrics; incomplete solution	12:16 PM Apr 22, 2024

Q.
1

▼ Question 1

🕒 Time taken: 20m 23s

Marks Scored: 1/4

Question

Please answer this question only by typing your answers in the typing space available for this question

The social media platform's data engineering team has observed that the data processing latency for user posts has increased by 20% compared to the previous month. [1 + 3 = 4 Marks]

- a) If the average processing latency for user posts in the previous month was 50 milliseconds, calculate the current average processing latency for user posts.
- b) Additionally, discuss the potential impact of this latency increase on the overall performance of the data pipeline and propose strategies to mitigate it.

Response

a) last month processing latency = 50 ms

given 20% increase in procesing latency in current month compare with last month

Current month latency = last month latency * (1+ % increase)

= 50 * (1+0.2) = 60 ms

The current month processing latency is 60 ms

b) Impacts of latency increase

- **Slowed down real time features** : Due to latency Features which depend on real-time data will be delayed and will impact on the user-experinece
- **Data pipeline backlogs**: Data will be backlogged in data pipeline which could results in further delay and may results in data loss also
- **Delay in analytics and reports**: Downstream analytics and reporting dashboards migh show outdated information due to slow in data prepartaion

Mitigation methods :

- **Identify Bottlenecks** : Identify the root cause for latency increase which could be from data-ingestion , transforamtion etc
- **Resource optimiztion** : Scaling up the needed resource like memory, CPUS for pre-processing tasks.
- **Code optimizatin** : optimize the data processing code for efficieny and adapt any proven algorithms or explore any new alogirthms which best suits for the data-processing
- **Paralleization**: Breakdown the large tasks in to multiple small tasks and enable a parellilization whcih can imrove the data processing
- **Data caching** : Implement any cachening mechnasim for frequently used data which will reduce the data processing time
- **horizontal scaling** : if resouce scaling is limited, consider adding more processing nodes to distribute the processing load horizontally
- **Microbatch processing** : data process could be done in a min-batch more frequently instead of large batches at slower speeds.

By adopting the above strategies which ever could be applicable for the current scenario it cold be possible to increase overall performace of data pipeline by decreasing the latency of the data processing

Words : 274

Max marks: 4

EVALUATOR'S NAME	MARKS ⚡	EVALUATION COMMENTS	LAST MODIFIED ON ⚡
Tharabai Baboo	1.0	a - Looks fine;b - copied	08:17 PM Apr 27, 2024

Test Log

14th Apr 2024

08:37 AM		Entered the waiting room
08:38 AM		Mobile phone paired - camera : camera2 1, facing front - microphone : Default
08:40 AM		Candidate gave us right to the following feeds - camera : USB Camera (0bda:57b5) - microphone : - Headset Microphone (Jabra EVOLVE LINK MS) (0b0e:0305)
08:40 AM		Mobile session has started
08:41 AM		Candidate Not Present
08:42 AM		Candidate Face not Visible
08:42 AM		Proctor(OKGProctor27) : Dear Candidate, we wish you all the best. I am your proctor. I will be monitoring you throughout the duration of the examination through your laptop /Desktop web camera and your mobile camera. In case you are facing any issues during the exam. Pls reach out to me for assistance
08:42 AM		Proctor(OKGProctor27) : Dear Candidate- Please note 1. You are not allowed to leave your seat during exam. 2. No one is permitted inside the room during the exam duration. 3. No Earphones /headphones, Ear pods, Bluetooth devices /Smartwatches are allowed. 4. No water break / snack break (kindly keep it handy) 5. Bio break will be allowed for 2- 3 minutes, kindly inform me and go 6. Any student found using any unfair means during the exam will be disqualified. 7. Ensure that your mobile at a height, placed to 2 /2.5 meters away, to be kept on a stack of books to give a better view. 8. Your Hands, Keyboard and Writing Area Should be visible all the time.
08:42 AM		Proctor(OKGProctor27) : Dear Candidate, please rotate your Mobilecam 360 degree and show the test environment at your place. 360-degree view means rotate the camera all around to have full circular view of the room / surroundings (top, bottom, left, right and floor) and the working table where your laptop/desktop is been placed.
08:42 AM		Candidate Not Present for 01 min
08:43 AM		Proctor(OKGProctor27) : Dear Candidate, Thankyou.
08:48 AM		Candidate : is usb mouse allowed. As this is typing exam
08:49 AM		Candidate : wireless USB mouse for laptop
08:50 AM		Proctor(OKGProctor27) : Dear candidate these are not allowed.
08:50 AM		Candidate : ok
08:50 AM		Proctor(OKGProctor27) : Dear Candidate, Thankyou.
08:51 AM		Candidate Face not Visible
08:51 AM		Candidate Looking Away from Screen
08:58 AM		Candidate Looking Away from Screen
09:00 AM		Started the test with Section #1
09:00 AM		Started the test in Mettl Secure Browser
09:00 AM		Went to Section #2 of the test
09:00 AM		Went to Section #3 of the test
09:00 AM		Went to Section #4 of the test
09:01 AM		Went to Section #5 of the test
09:01 AM		Went to Section #4 of the test

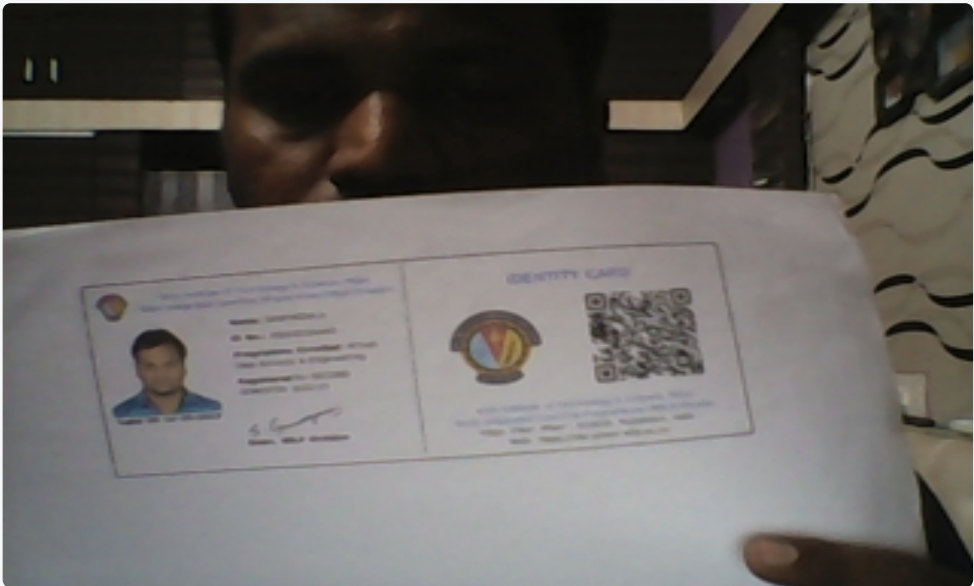
09:01 AM	●	Went to Section #3 of the test
09:01 AM	●	Went to Section #2 of the test
09:02 AM	●	Went to Section #1 of the test
09:15 AM	●	Proctor(OKGSupervisor21) [Announcement] : Dear Candidate, This is a typing-only exam. Please type your answers in the textbox provided below each question. There will be NO Scan & upload option for this course and handwritten answers on answer scripts will NOT be considered for evaluation
09:16 AM	●	Went to Section #2 of the test
09:19 AM	●	Went to Section #3 of the test
09:36 AM	●	Went to Section #4 of the test
09:51 AM	●	Candidate Looking Away from Screen
09:53 AM	●	Proctor(OKGSupervisor21) [Announcement] : Dear Candidate, This is a typing-only exam. Please type your answers in the textbox provided below each question. There will be NO Scan & upload option for this course and handwritten answers on answer scripts will NOT be considered for evaluation
09:59 AM	●	Candidate Looking Away from Screen
10:00 AM	●	Went to Section #5 of the test
10:00 AM	●	Went to Section #4 of the test
10:01 AM	●	Went to Section #3 of the test
10:01 AM	●	Went to Section #4 of the test
10:01 AM	●	Went to Section #5 of the test
10:01 AM	●	Went to Section #4 of the test
10:01 AM	●	Went to Section #3 of the test
10:02 AM	●	Went to Section #2 of the test
10:02 AM	●	Went to Section #1 of the test
10:03 AM	●	Went to Section #2 of the test
10:04 AM	●	Went to Section #1 of the test
10:04 AM	●	Went to Section #2 of the test
10:05 AM	●	Went to Section #3 of the test
10:05 AM	●	Went to Section #2 of the test
10:05 AM	●	Went to Section #1 of the test
10:05 AM	●	Went to Section #2 of the test
10:07 AM	●	Went to Section #1 of the test
10:07 AM	●	Went to Section #2 of the test
10:08 AM	●	Went to Section #1 of the test
10:08 AM	●	Went to Section #2 of the test
10:09 AM	●	Went to Section #3 of the test
10:17 AM	●	Went to Section #4 of the test
10:17 AM	●	Went to Section #3 of the test
10:17 AM	●	Went to Section #2 of the test
10:17 AM	●	Went to Section #1 of the test
10:17 AM	●	Went to Section #2 of the test

10:17 AM	●	Went to Section #3 of the test
10:18 AM	●	Went to Section #4 of the test
10:18 AM	●	Went to Section #5 of the test
10:18 AM	●	Went to Section #4 of the test
10:18 AM	●	Went to Section #3 of the test
10:18 AM	●	Went to Section #2 of the test
10:43 AM	●	Went to Section #3 of the test
10:43 AM	●	Went to Section #4 of the test
10:44 AM	●	Went to Section #5 of the test
11:02 AM	●	Went to Section #4 of the test
11:02 AM	●	Went to Section #5 of the test
11:02 AM	●	Went to Section #4 of the test
11:02 AM	●	Went to Section #3 of the test
11:02 AM	●	Went to Section #2 of the test
11:02 AM	●	Went to Section #1 of the test
11:14 AM	●	Went to Section #2 of the test
11:14 AM	●	Went to Section #3 of the test
11:18 AM	●	Went to Section #4 of the test
11:25 AM	●	Went to Section #5 of the test
11:27 AM	●	Went to Section #4 of the test
11:27 AM	●	Went to Section #3 of the test
11:30 AM	●	Mobile session has ended
11:30 AM	🚩	Test ended due to time over

Profile Picture Snapshot



Identity Card Snapshot



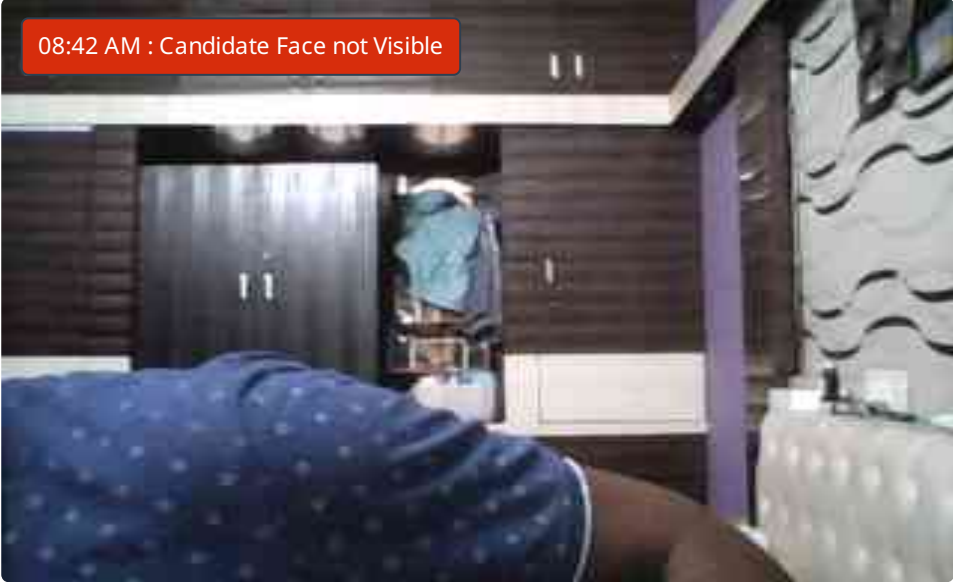
08:41 AM : Candidate Not Present



08:41 AM : Candidate Not Present



08:42 AM : Candidate Face not Visible



08:42 AM : Candidate Not Present



08:43 AM : Candidate Not Present



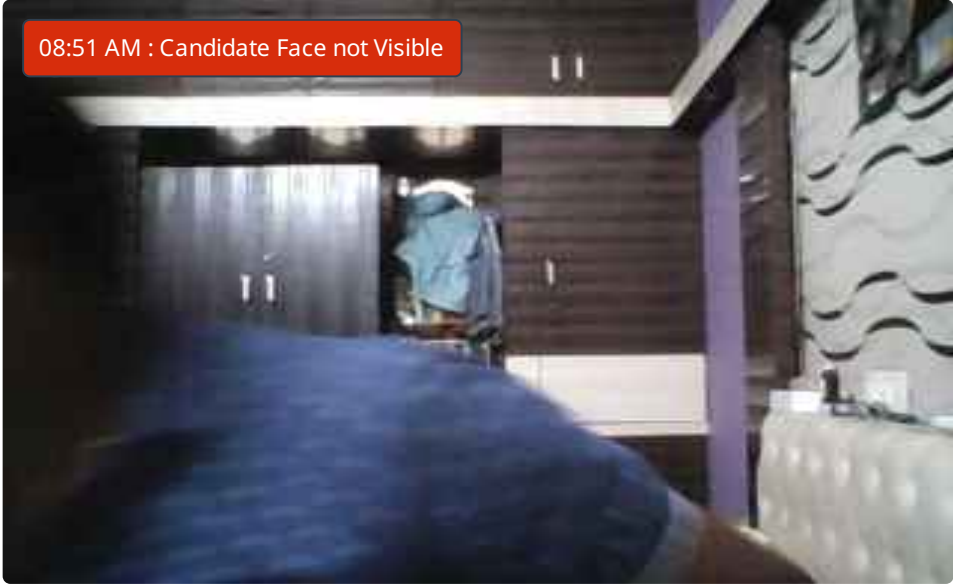
08:43 AM : Candidate Not Present

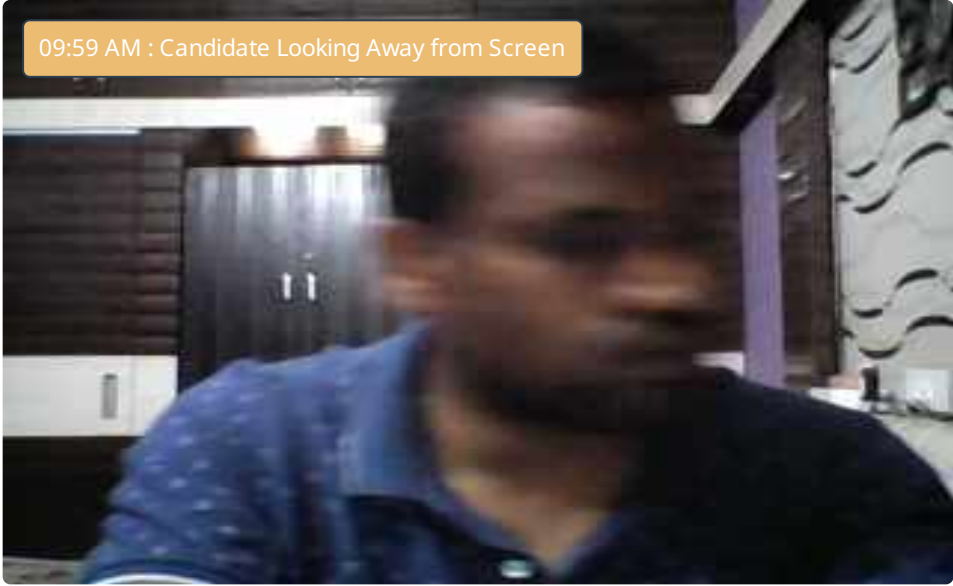
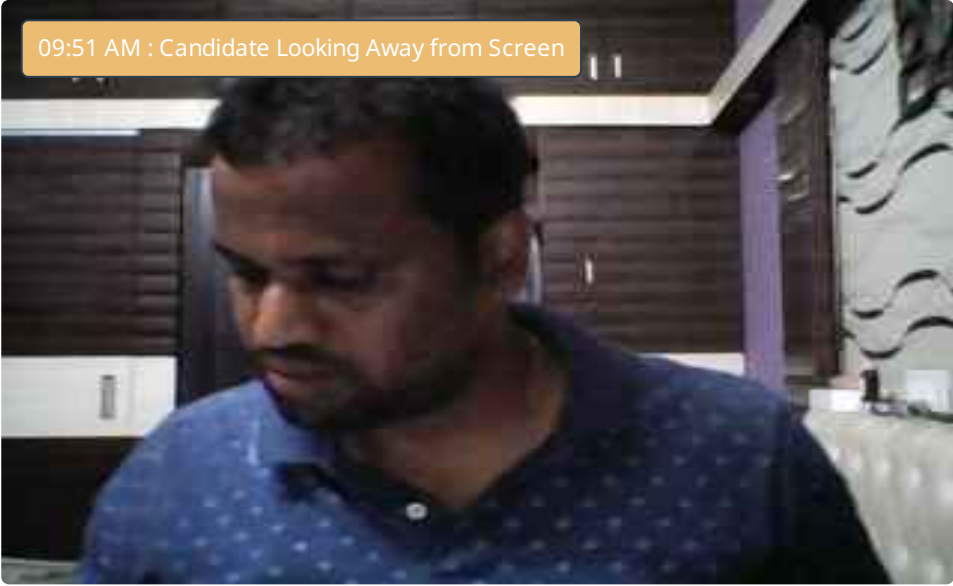
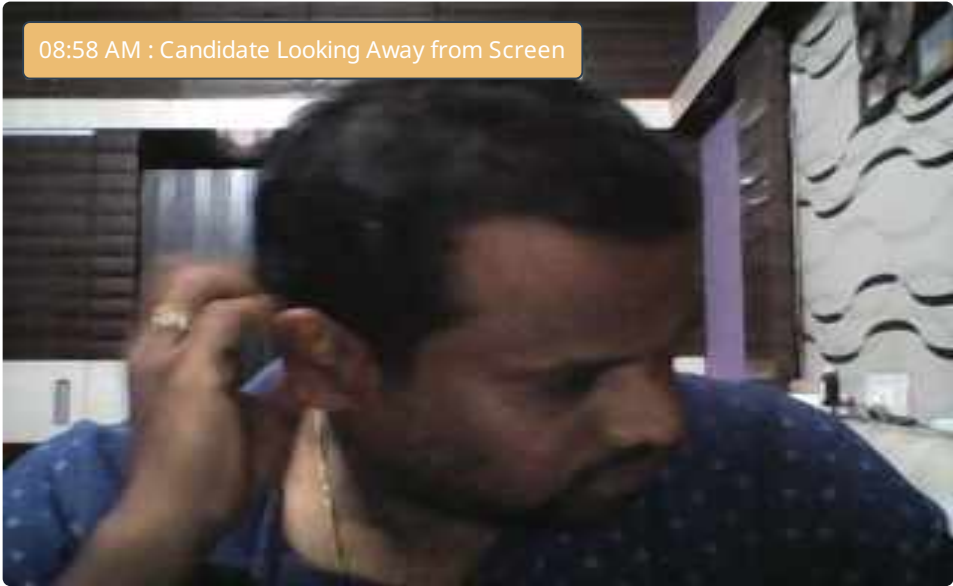
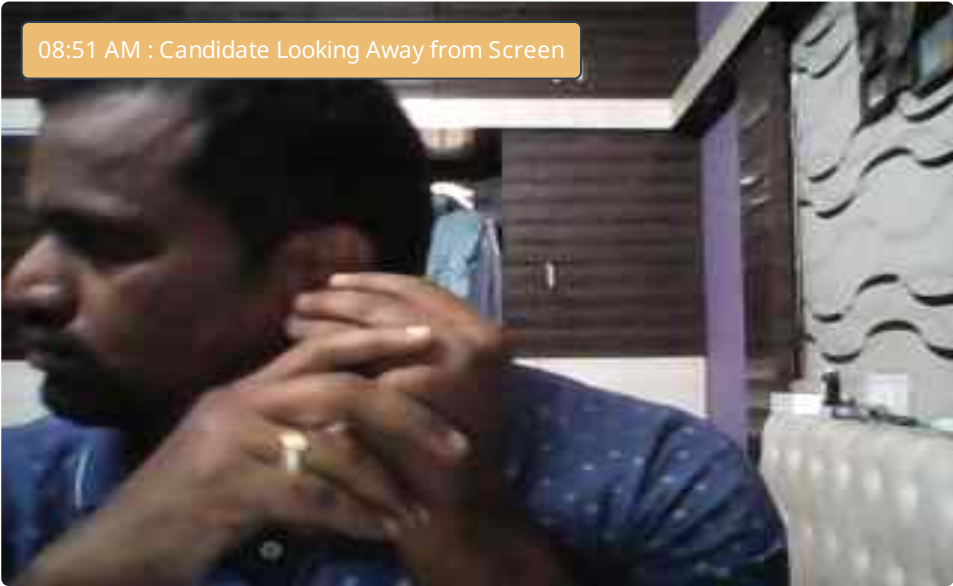


08:43 AM : Candidate Not Present



08:51 AM : Candidate Face not Visible





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